

SDLC Model For AutoApply

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Use Case model for AutoApply :

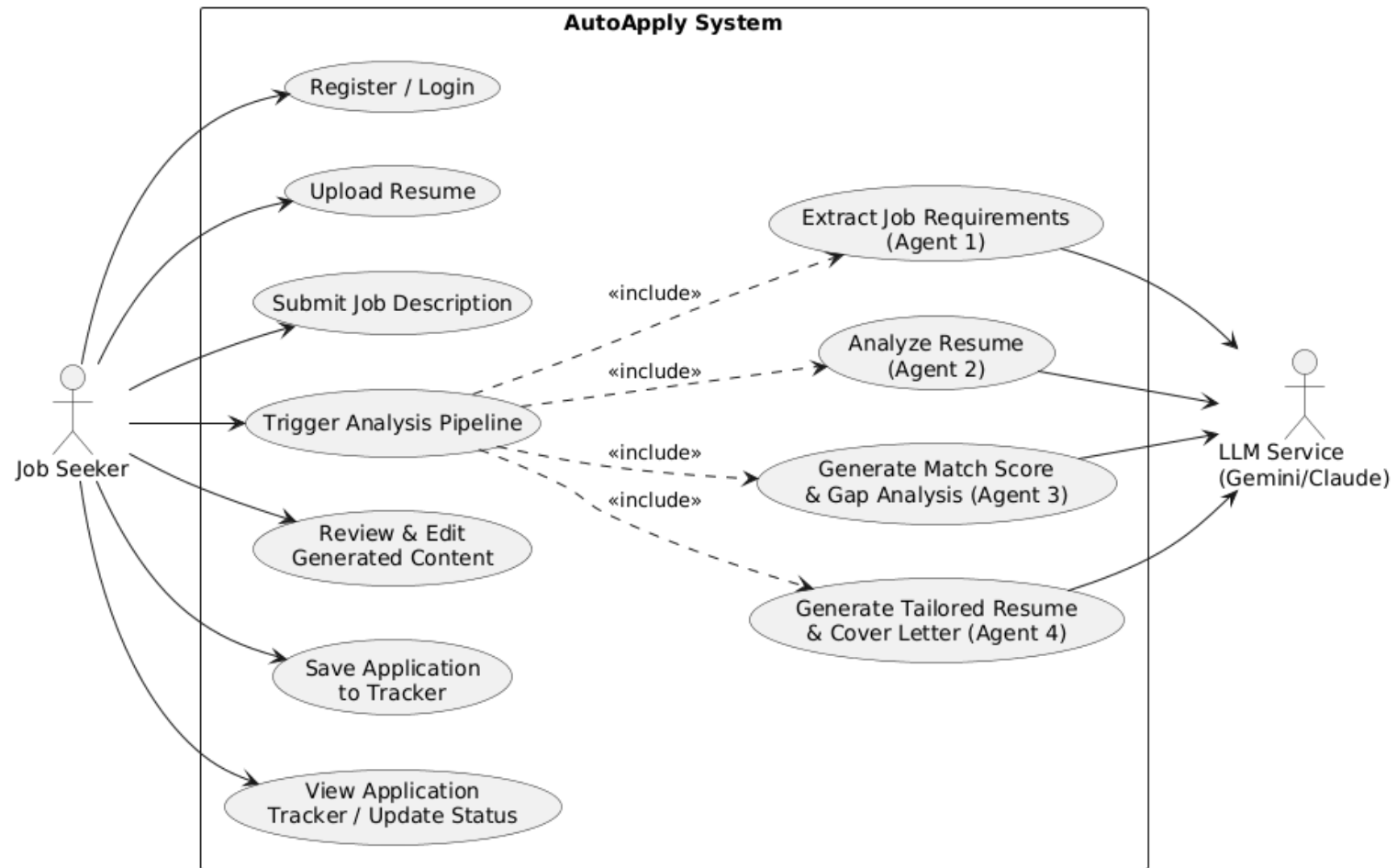


Table 1: Comparison Matrix with Different Models

Priority	Criteria	Waterfall	V-Shaped	Iterative	Spiral	Agile	Prototype
5	Requirement Flexibility	No	No	Yes	Yes	Yes	Yes
4	AI Output Adaptability (Prompt/Response Tuning)	No	No	Yes	No	Yes	Yes
6	Continuous Testing & Integration Feedback	No	Yes	Yes	Yes	Yes	No
4	Risk Management & AI Reliability Handling	No	No	No	Yes	Yes	No
5	User Feedback & Early Delivery	No	No	Yes	Yes	Yes	Yes
3	Efficiency & Low Overhead	Yes	No	Yes	No	Yes	No
3	Fast Delivery (Short Time-to-Market)	No	No	Yes	No	Yes	Yes
-	Total (Priority Weights)	30					
	Overall Score	3	6	23	20	30	17

Justification:

Agile scored highest (30/30) because AutoApply's core component — the four-agent AI pipeline — requires iterative development and continuous testing. Each agent (job parser, resume analyzer, matcher, writer) is built and tested independently, then integrated in stages, matching Agile's sprint-based approach. Since the system uses pasted job descriptions rather than live web scraping, the main risk is AI output reliability (inconsistent or malformed responses from the language model) rather than bot detection. AI-generated output is not fully predictable in advance, so requirements naturally evolve as real outputs are reviewed — Agile supports this flexibility better than upfront-heavy models like Waterfall or V-Shape. Early, incremental delivery (single agent → 2-agent chain → full pipeline → frontend → tracker) also aligns directly with Agile's principle of frequent working increments.